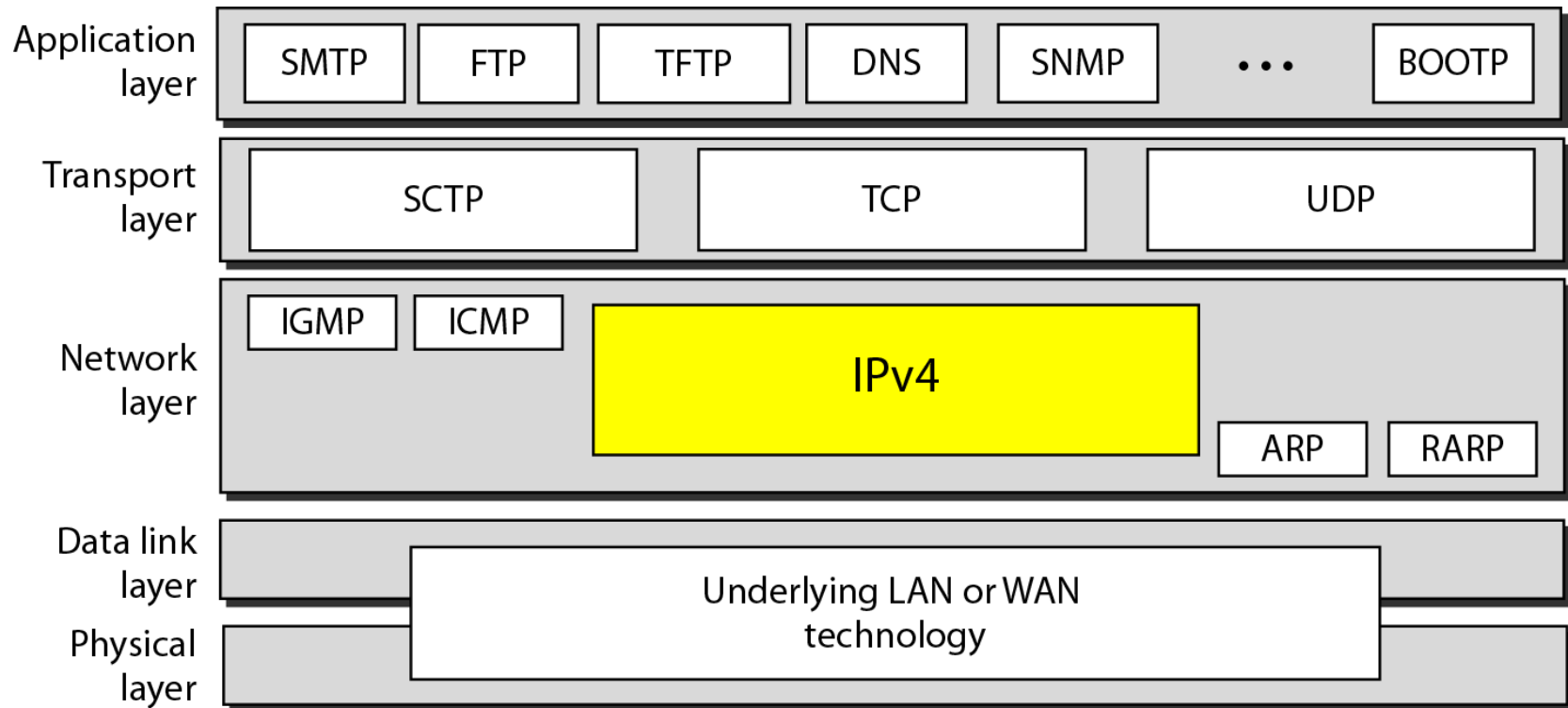


Internet Protocol

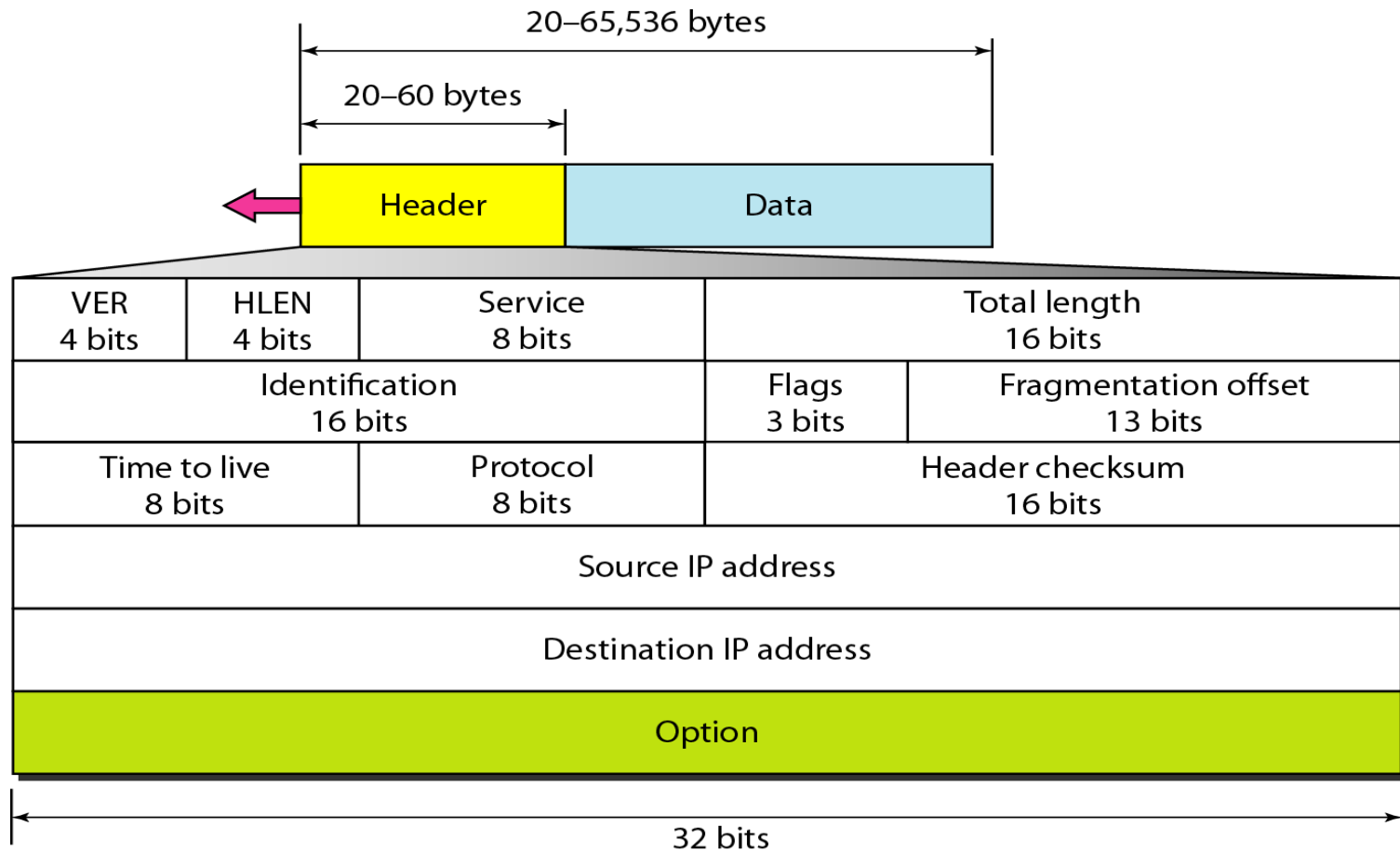
- IP is an unreliable and connectionless datagram protocol
-a **best-effort delivery service**.
- Best-effort means that IP packets can be corrupted, lost, arrive out of order, or delayed and may create congestion for the network.
- If reliability is important, IP must be paired with a reliable protocol such as TCP

Position of IPv4 in TCP/IP protocol suite



IPv4 datagram

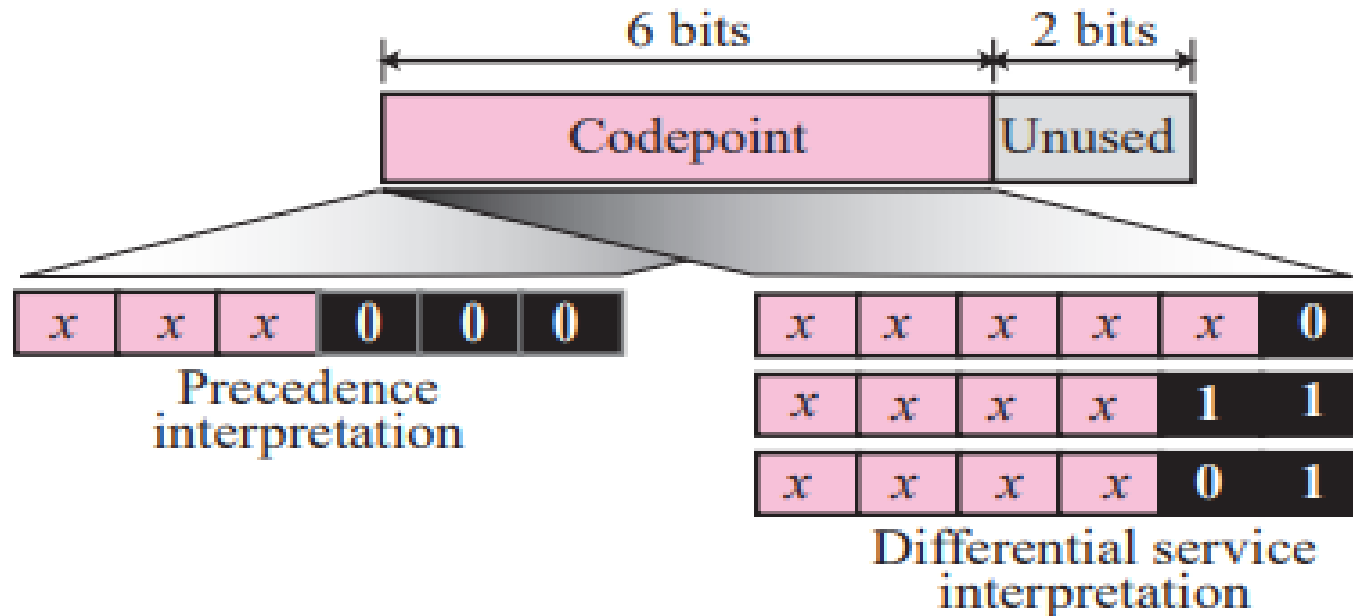
(Packets in the network (internet) layer are called **datagrams**)



IPv4 Datagram Format

- Version (VER): version of the IP protocol. Currently, the version is 4.
- Header length (HLEN): the total length of the datagram header in 4-byte words.
- Services: service type or differentiated services (not used now).
- Total length: total length (header plus data) of the datagram in bytes.
 - $\text{Total length of data} = \text{total length} - \text{header length}$

Service type



Values for codepoints

Category	Codepoint	Assigning Authority
1	XXXXX0	Internet
2	XXXX11	Local
3	XXXX01	Temporary or experimental

IPv4 Datagram Format

- Identification: used in fragmentation (discussed later).
- Flags: used in fragmentation (discussed later).
- Fragmentation offset: used in fragmentation (discussed later).
- Time to live: it is used to control the maximum number hops visited by the datagram.
- Protocol: defines the higher-level protocol that uses the services of the IPV4 layer.

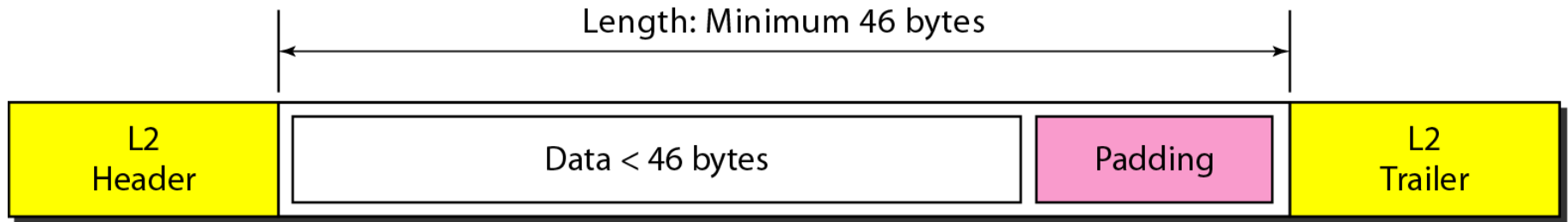
IPv4 Datagram Format

- Checksum: 1's compliment checksum (introduced in Chapter 10).
- Source address: is the IPv4 address of the source.
- Destination address: is the IPv4 address of the source.

Note

The total length field defines the total length of the datagram including the header.

Figure 20.7 *Encapsulation of a small datagram in an Ethernet frame*



**One of the reason why “total length”
field is required.**

Figure 20.8 *Protocol field and encapsulated data*

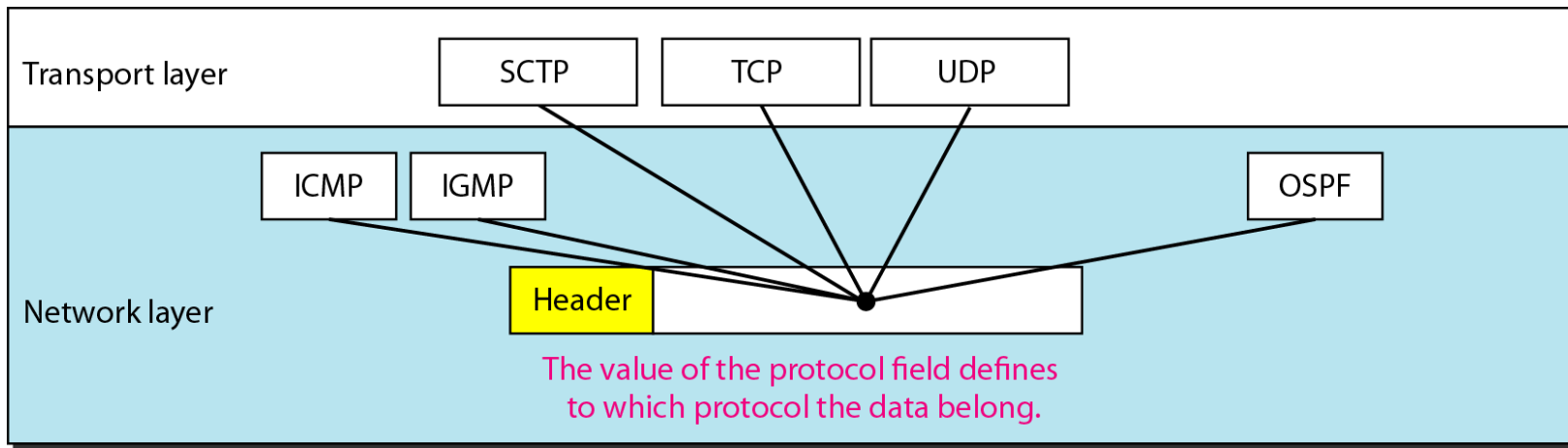


Table 20.4 *Protocol values*

<i>Value</i>	<i>Protocol</i>
1	ICMP
2	IGMP
6	TCP
17	UDP
89	OSPF

Figure 20.9 *Maximum transfer unit (MTU)*

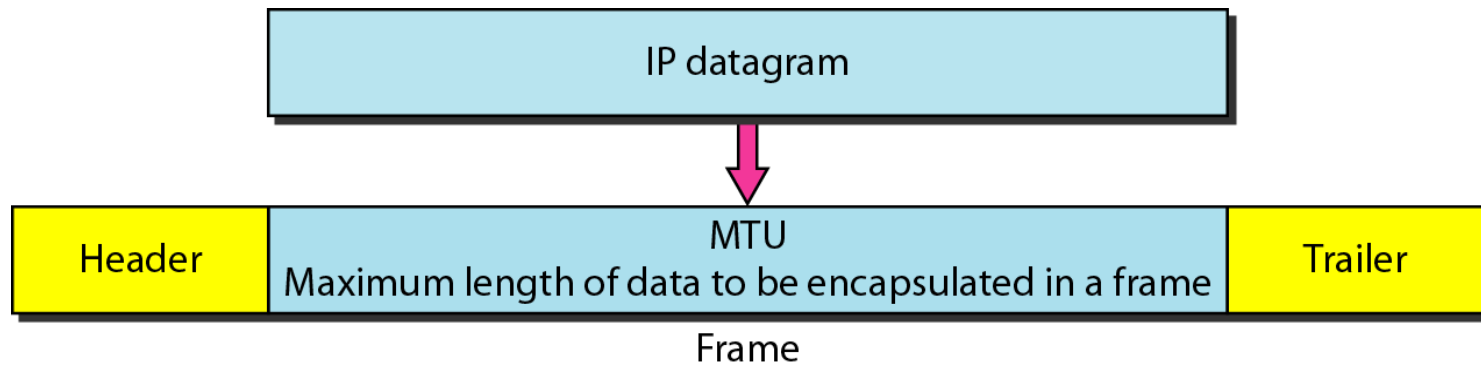


Table 20.5 *MTUs for some networks*

<i>Protocol</i>	<i>MTU</i>
Hyperchannel	65,535
Token Ring (16 Mbps)	17,914
Token Ring (4 Mbps)	4,464
FDDI	4,352
Ethernet	1,500
X.25	576
PPP	296

Fields Related to Fragmentation

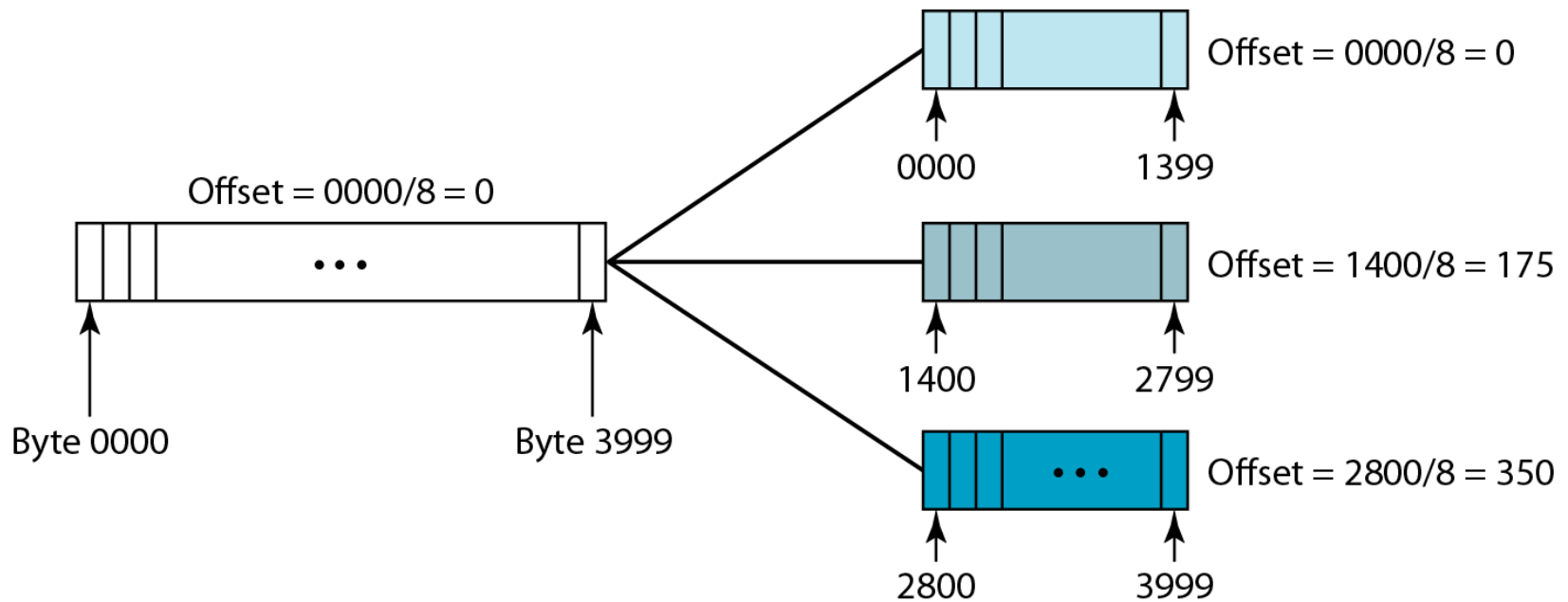
- **Identification:** identifies a datagram originating from the source host. A combination of the identification and source address must uniquely define a datagram as it leaves the source node.
- **Flags:** see next slide.
- **Fragmentation offset:** is the offset of the data in the original datagram measured in units of 8 bytes.

Figure 20.10 *Flags (3 bits) used in fragmentation*



- first bit: reserved (not used)
 - second bit: = 1 requires the packet not to be fragmented
drops the packet if it is $>$ MTU
 - third bit: =1 more fragmented packets later
=0 the last fragmented packet
-

Figure 20.11 *Fragmentation example*



IP Fragmentation and Reassembly

Example

- ❑ 4000 byte datagram
- ❑ MTU = 1500 bytes

	length	ID	fragflag	offset
	=4000	=x	=0	=0

One large datagram becomes several smaller datagrams

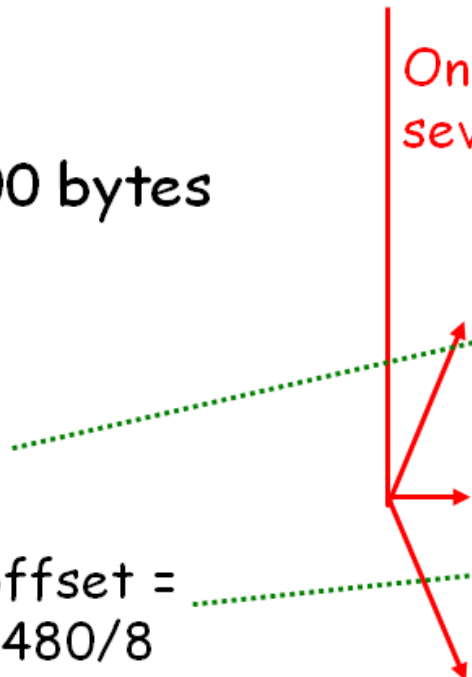
	length	ID	fragflag	offset
	=1500	=x	=1	=0

	length	ID	fragflag	offset
	=1500	=x	=1	=185

	length	ID	fragflag	offset
	=1500 =1060	=x	=0	=370

1480 bytes in
data field

offset =
 $1480/8$



Example 19.6

A packet has arrived with an M bit value of 0. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

If the M bit is 0, it means that there are no more fragments; the fragment is the last one. However, we cannot say if the original packet was fragmented or not. A nonfragmented packet is considered the last fragment.

Example 19.7

A packet has arrived with an M bit value of 1. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

If the M bit is 1, it means that there is at least one more fragment. This fragment can be the first one or a middle one, but not the last one. We don't know if it is the first one or a middle one; we need more information (the value of the fragmentation offset).

Example 19.8

A packet has arrived with an M bit value of 1 and a fragmentation offset value of 0. Is this the first fragment, the last fragment, or a middle fragment?

Solution

Because the M bit is 1, it is either the first fragment or a middle one. Because the offset value is 0, it is the first fragment.

Example 19.9

A packet has arrived in which the offset value is 100. What is the number of the first byte? Do we know the number of the last byte?

Solution

To find the number of the first byte, we multiply the offset value by 8. This means that the first byte number is 800. We cannot determine the number of the last byte unless we know the length of the data.

Example 19.10

A packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?

Solution

The first byte number is $100 \times 8 = 800$. The total length is 100 bytes, and the header length is 20 bytes (5×4), which means that there are 80 bytes in this datagram. If the first byte number is 800, the last byte number must be 879.



IPv4 Checksum

- *IPv4 checksum use the 1's compliment method (chapter 10)*
- *Checksum only computes for IP header, not data*
 - *Upper layer has checksum for data portion*
 - *Header always changes in each router*
- *Header is chunked to 16-bit sections for computing*

Figure 20.13 *Example of checksum calculation in IPv4*

4	5	0	28			
1			0	0		
4		17	0			
10.12.14.5						
12.6.7.9						

4, 5, and 0

28

1

0 and 0

4 and 17

0

10.12

14.5

12.6

7.9

Sum

Checksum

→

→

→

→

→

→

→

→

→

→

→

→

4

5

0

0

0

0

0

A

E

C

7

4

4

8

B

B

1

0

0

0

0

1

0

0

0

0

5

6

9

E

IPv6 ADDRESS

Despite all short-term solutions, address depletion is still a long-term problem for the Internet. This and other problems in the IP protocol itself have been the motivation for IPv6.



Note

An IPv6 address is 128 bits long.

Figure 19.14 *IPv6 address in binary and hexadecimal colon notation*

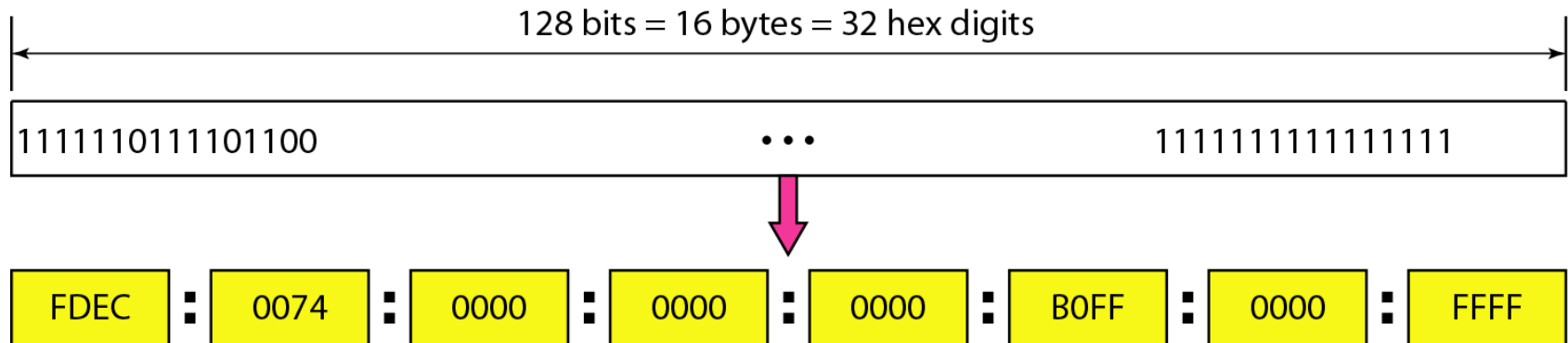
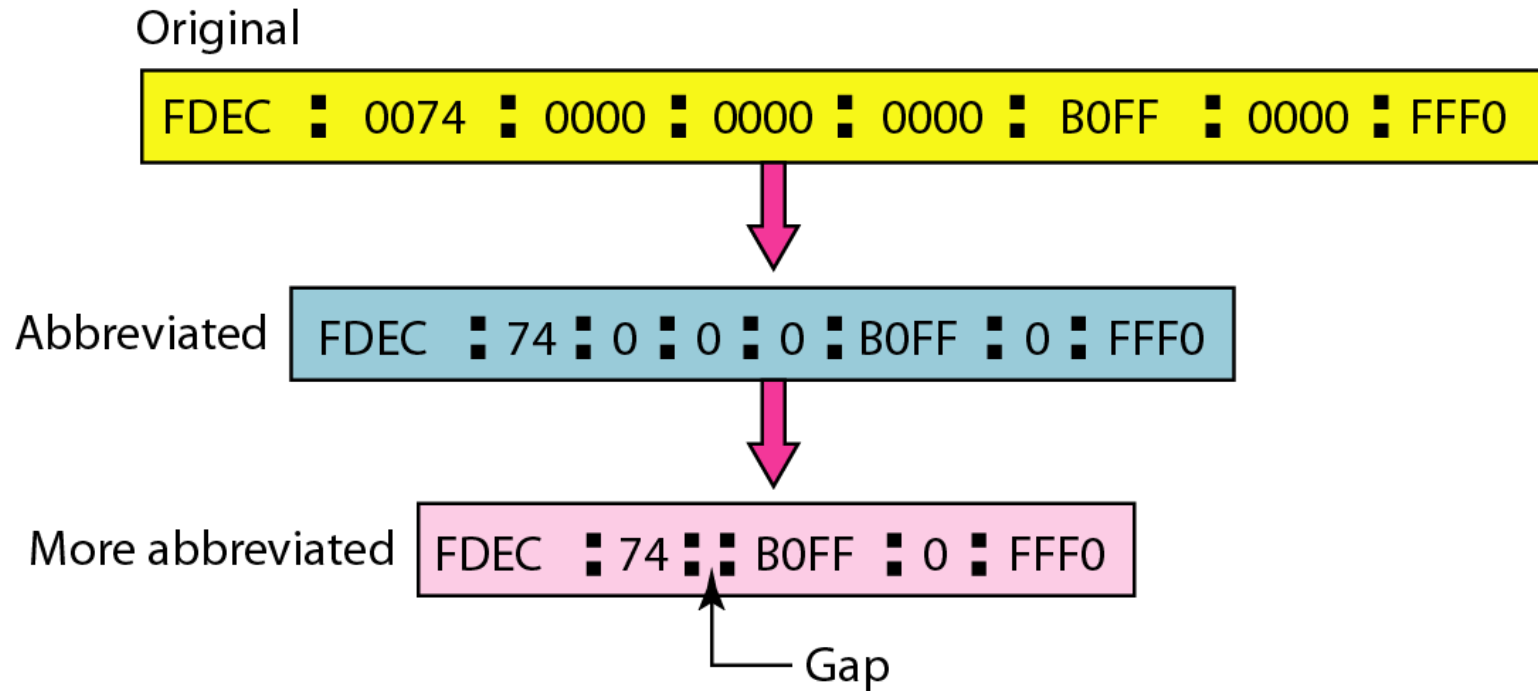


Figure 19.15 *Abbreviated IPv6 addresses*



IPv6 Datagram

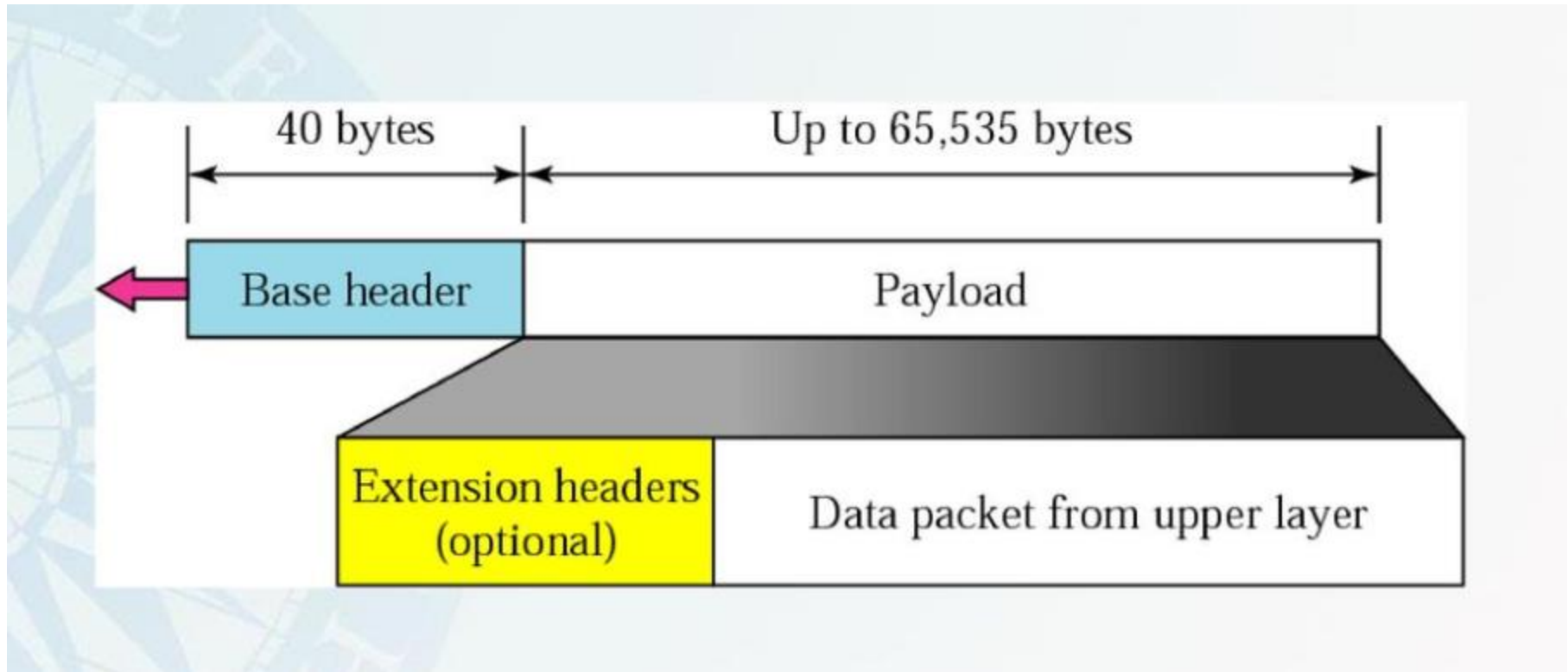


Figure 20.16 *Format of an IPv6 datagram*

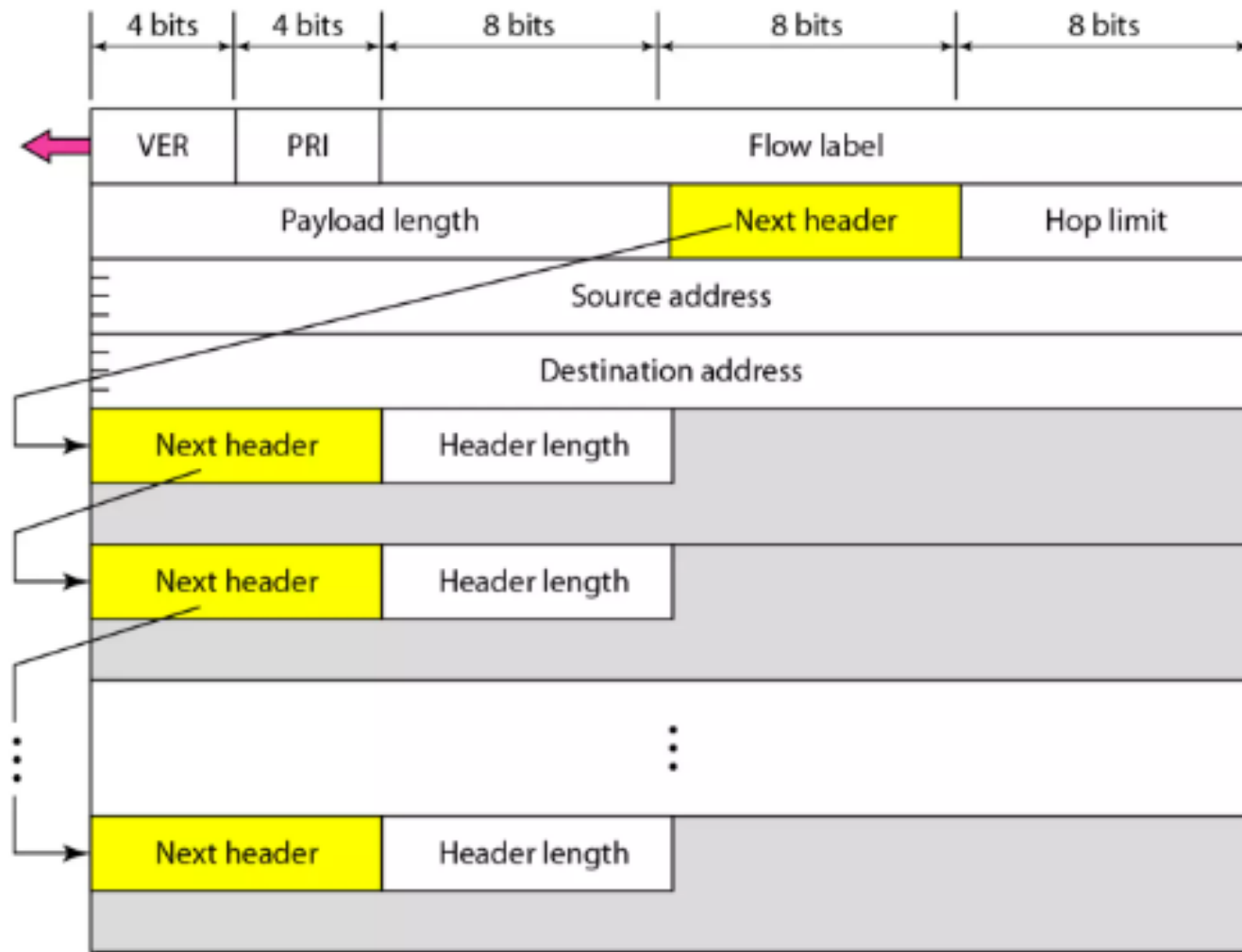


Table 20.7 *Priorities for congestion-controlled traffic*

<i>Priority</i>	<i>Meaning</i>
0	No specific traffic
1	Background data
2	Unattended data traffic
3	Reserved
4	Attended bulk data traffic
5	Reserved
6	Interactive traffic
7	Control traffic

Table 20.8 *Priorities for noncongestion-controlled traffic*

<i>Priority</i>	<i>Meaning</i>
8	Data with greatest redundancy
...	...
15	Data with least redundancy

IPv6 Header

Version	Traffic Class	Flow Label	
Payload Length		Next Header	Hop Limit
Source Address			
Destination Address			

IPv4 Header

Version	IHL	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol	Header Checksum		
Source Address				
Destination Address				
Options				Padding

Legend



Fields **kept** in IPv6



Fields **kept** in IPv6, but name and position changed



Fields **not kept** in IPv6



Fields that are **new** in IPv6

Comparison

1. The header length field is eliminated in IPv6 because the length of the header is fixed in this version.
2. The service type field is eliminated in IPv6. The priority and flow label fields together take over the function of the service type field.
3. The total length field is eliminated in IPv6 and replaced by the payload length field.
4. The identification, flag, and offset fields are eliminated from the base header in IPv6. They are included in the fragmentation extension header.
5. The TTL field is called hop limit in IPv6.
6. The protocol field is replaced by the next header field.
7. The header checksum is eliminated because the checksum is provided by upper-layer protocols; it is therefore not needed at this level.
8. The option fields in IPv4 are implemented as extension headers in IPv6.

Property	IPv4	IPv6
Address size and network size	32 bits, network size 8-30 bits	128 bits, network size 64 bits
Packet header size	20-60 bytes	40 bytes
Header-level extension	limited number of small IP options	unlimited number of IPv6 extension headers
Fragmentation	sender or any intermediate router allowed to fragment	only sender may fragment
Control protocols	mixture of non-IP (ARP), ICMP, and other protocols	all control protocols based on ICMPv6
Minimum allowed MTU	576 bytes	1280 bytes
Path MTU discovery	optional, not widely used	strongly recommended
Address assignment	usually one address per host	usually multiple addresses per interface
Address types	use of unicast, multicast, and broadcast address types	broadcast addressing no longer used, use of unicast, multicast and anycast address types
Address configuration	devices configured manually or with host configuration protocols like DHCP	devices configure themselves independently using stateless address autoconfiguration (SLAAC) or use DHCP